

ACCT102 Management Accounting

Week 5

Service department allocation

Service and Operating Departments

Service Department

- Do not directly engage in operating activities

Examples:

- *Human Resource in a Hospital*
- *Accounts dept in a University*
- *Cafeteria in a Manufacturing firm*

Operating Department

- Carry out central purposes of organization

Examples:

- *Surgery dept in a Hospital*
- *School of Accountancy dept in a University*
- *Food production dept in a Manufacturing firm*

Reasons for Charging Service Department Costs



Costs of the service department become overhead costs to the operating department

Why do we allocate service department cost to operating departments?

1. To provide operating departments with more complete cost data for making decisions and measuring profitability
2. To encourage operating departments to wisely use service department resources.
3. To create an incentive for service departments to operate efficiently.

Factors to be considered in Cost Allocation Approach and Intercompany/Interdepartmental charges

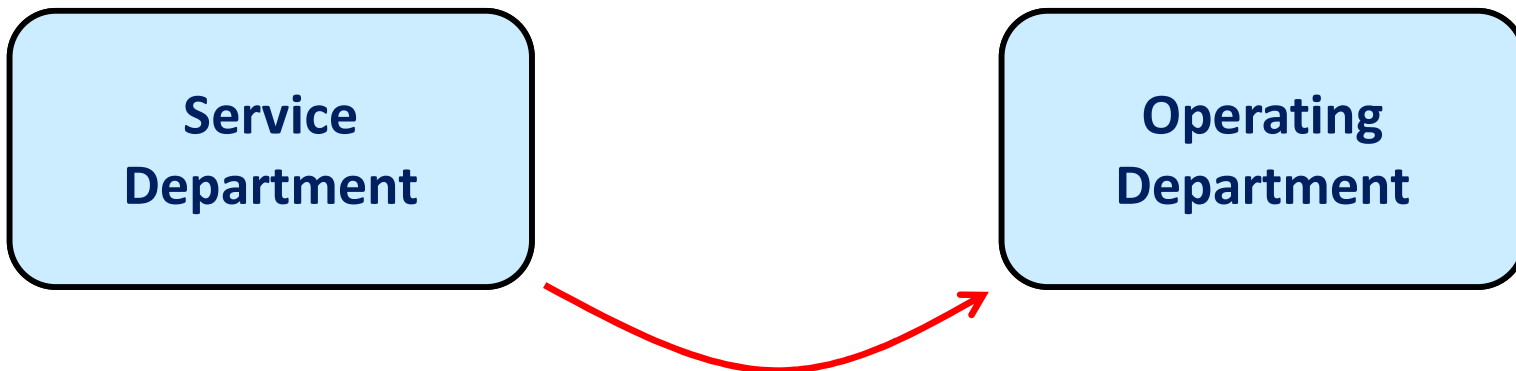
Cause and effect	Absorption of cost based on who causes it
Benefit Received	Absorption of cost based on the ultimate benefactor
Ability to Bear	Absorption of cost based on who/which product has the ability and profit margin to bear the cost
Fairness or Equity	Ensuring fair game and decent profit to motivate service provider, for example, use of a cost-plus performance-based award fee approach, to ensure quality delivery of service/product
Cost-benefit Evaluation	Including tangible and intangible costs and benefits

Learning Objective

Allocate costs of service departments to other operating departments/units using the cost behavior concept.

Charging Costs by Behavior

- If possible, variable and fixed service department costs should be charged to operating departments **separately**.
- **Budgeted (not actual)** variable and fixed service department costs should be charged to operating departments.



Costs of the service department become overhead costs to the operating department

Variable Service Department Costs

- Charge according to whatever **activity (or cost driver)** causes the incurrence of the cost.

Company cafeteria



Costs – ingredients
Activity – headcount

Maintenance



Costs – grease
Activity – machine hours

Fixed Service Department Costs

- Fixed costs of service departments **represent the costs of having a certain level of service capacity available and online** for each operating department.
- Charge in predetermined lump-sum amounts that are based on the consuming department's **peak-period or long-run average servicing needs**.
 - Should not vary from period to period.

Distribution



Costs - depreciation of trucks

Charging Costs by Behavior – Example (1)

Sipco has a maintenance service department and two operating departments: Cutting and Assembly.

Variable maintenance costs driven by machine hours are budgeted at **\$75,000**.
Fixed maintenance costs are budgeted at **\$200,000 per year**.

Operating Departments	Percent of Peak-Period Capacity Required	Hours Planned	Hours Used
Cutting	60%	75,000	80,000
Assembly	40%	50,000	40,000
Total hours	100%	125,000	120,000

Allocate maintenance costs to the two operating departments.

Charging Costs by Behavior – Example (2)

Operating Departments	Percent of Peak-Period Capacity Required	Hours Planned	Hours Used
Cutting	60%	75,000	80,000
Assembly	40%	50,000	40,000
Total hours	100%	125,000	120,000

Var: \$75k, driven by MH

Fixed: \$200k per year

Charging Costs by Behavior – Example (3)

Variable cost rate: \$0.60 per MH (\$125k / 75k hours)

Allocate based on **actual hours**

	<u>Cutting Department</u>	<u>Assembly Department</u>
Variable cost allocation:		
\$0.60 × 80,000 hours	\$ 48,000	
\$0.60 × 40,000 hours		\$ 24,000
Fixed cost allocation:		
60% × \$200,000	120,000	
40% × \$200,000		80,000
Total allocated cost	<u>\$ 168,000</u>	<u>\$ 104,000</u>

Allocate based on **peak-period capacity**.

Quick Check 5B.1

- Foster City has an ambulance service that is used by the two public hospitals.
- Variable ambulance costs are budgeted at \$134,400 per year.
- Fixed ambulance costs are budgeted at \$120,000 per year.

Hospitals	Percent of Peak-Period Capacity Required	Miles Planned	Miles Used
Mercy	45%	15,000	16,000
Northside	55%	17,000	17,500
Total	100%	32,000	33,500

How much VARIABLE ambulance service cost will be allocated to Mercy Hospital at the **end** of the year?

- a. \$67,200 c. \$63,000
b. \$73,500 d. \$71,400

Quick Check 5B.2

- Foster City has an ambulance service that is used by the two public hospitals.
- Variable ambulance costs are budgeted at \$134,400 per year.
- Fixed ambulance costs are budgeted at \$120,000 per year.

Hospitals	Percent of Peak-Period Capacity Required	Miles Planned	Miles Used
Mercy	45%	15,000	16,000
Northside	55%	17,000	17,500
Total	100%	32,000	33,500

How much FIXED ambulance service cost will be allocated to Northside Hospital at the **end** of the year?

- a. \$66,000 c. \$62,687
b. \$63,750 d. \$65,625

Pitfalls in Allocating Fixed Costs

Pitfall	Result
Allocating fixed costs using a variable allocation base that fluctuates period to period.	Fixed costs allocated to one department are heavily influenced by what happens in other departments.
Using sales dollars as an allocation base.	Sales of one department influence the service department costs allocated to other departments.

Pitfalls in Allocating Fixed Costs – Example

Autos R Us has 1 service department and 2 sales departments, New Cars, Used Cars. Contrary to good practice, Autos R Us allocates the service department costs based on sales.

YEAR 1		Departments	
	Total	New	Used
Sales	\$ 1,000,000	\$ 500,000	\$ 500,000
<i>% of sales</i>		50%	50%
Service dept fixed cost	\$ 200,000	\$ 100,000	\$ 100,000

YEAR 2		Departments	
	Total	New	Used
Sales	\$ 1,250,000	\$ 750,000	\$ 500,000
<i>% of sales</i>		60%	40%
Service dept fixed cost	\$ 200,000	\$ 120,000	\$ 80,000

As the manager of the New Cars department,

- Service department capacity (fixed cost) remains the same.
- You did well in Y2, increasing sales by 50%
- Resulted in attracting 20% more fixed cost allocated

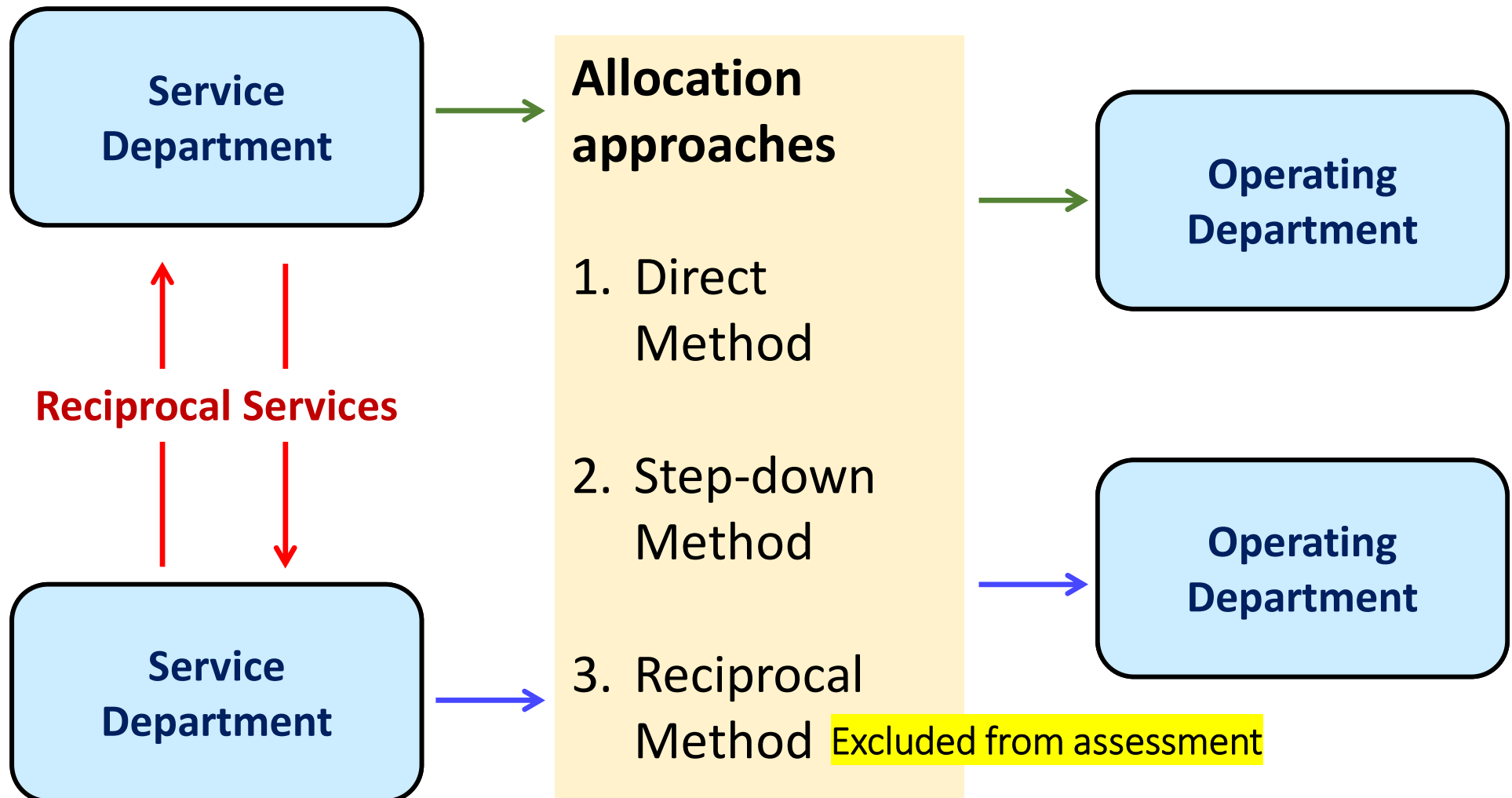
Would you complain?

Learning Objective

Allocate service department costs to operating departments using the:

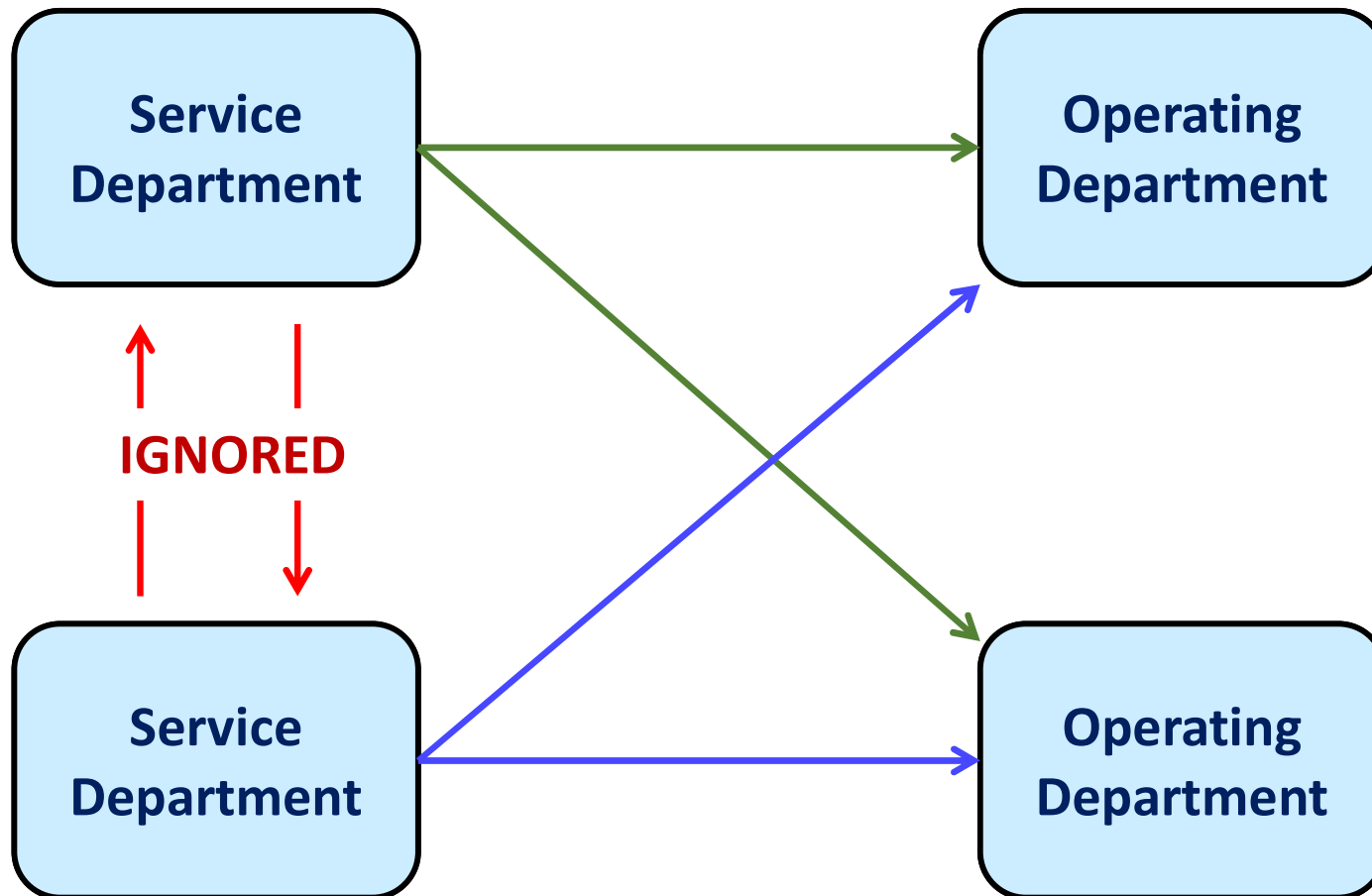
- (1) direct method,
- (2) step-down method,
- (3) reciprocal method. Excluded from assessment

Service Department Cost Allocations Approaches



Direct Method

Interactions between service departments are ignored and all costs are allocated directly to operating departments.



Direct Method – Example

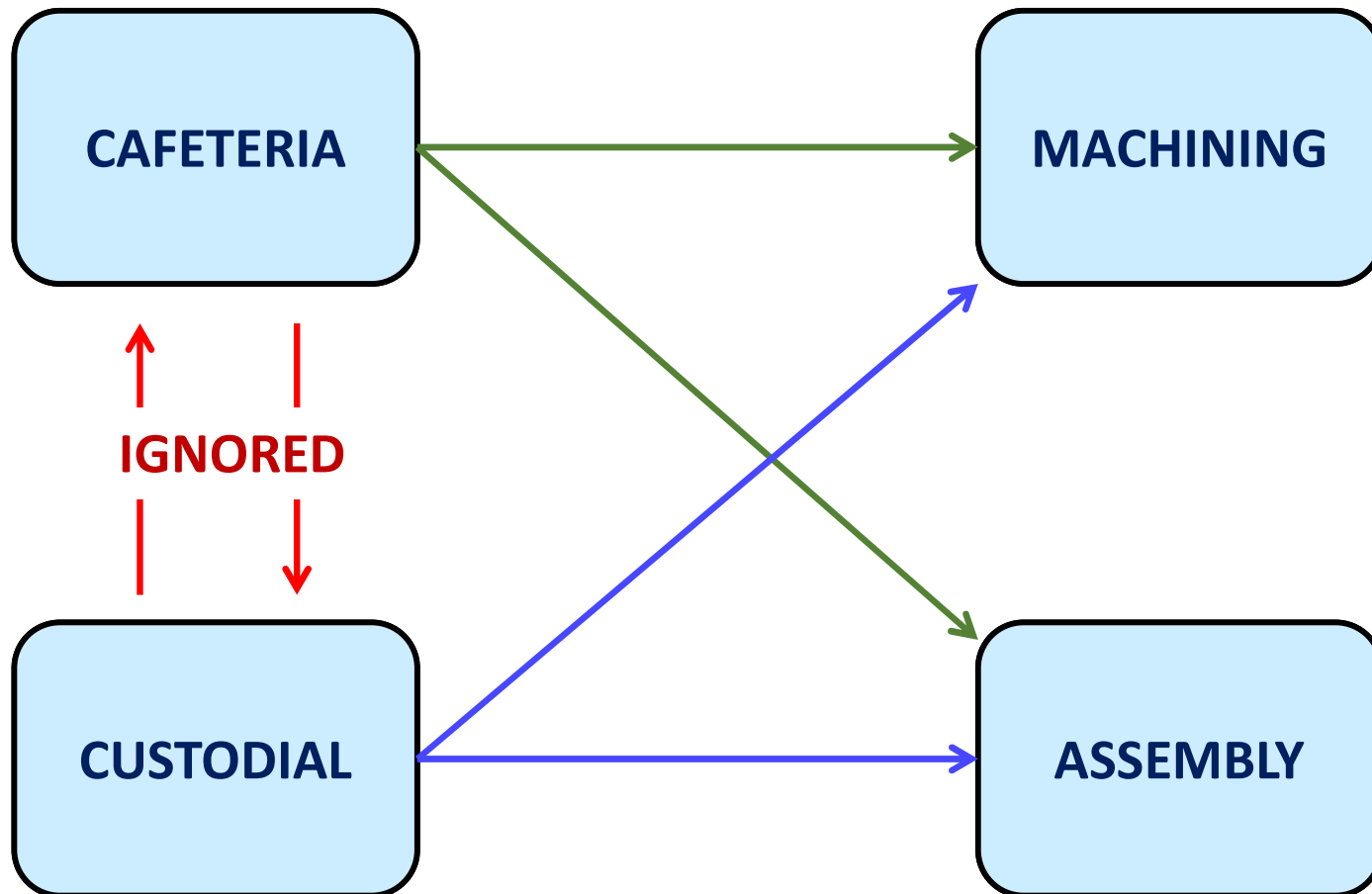
How much of the Cafeteria and Custodial costs should be allocated to each operating department using the direct method of cost allocation?

	<u>Service Departments</u>		<u>Operating Departments</u>	
	<u>Cafeteria</u>	<u>Custodial</u>	<u>Machining</u>	<u>Assembly</u>
Departmental costs before allocation	\$ 360,000	\$ 90,000	\$ 400,000	\$ 700,000
Number of employees	15	10	20	30
Square feet occupied	5,000	2,000	25,000	50,000

<u>Service Department</u>	<u>Allocation Base</u>
Cafeteria	Number of employees
Custodial	Square feet occupied

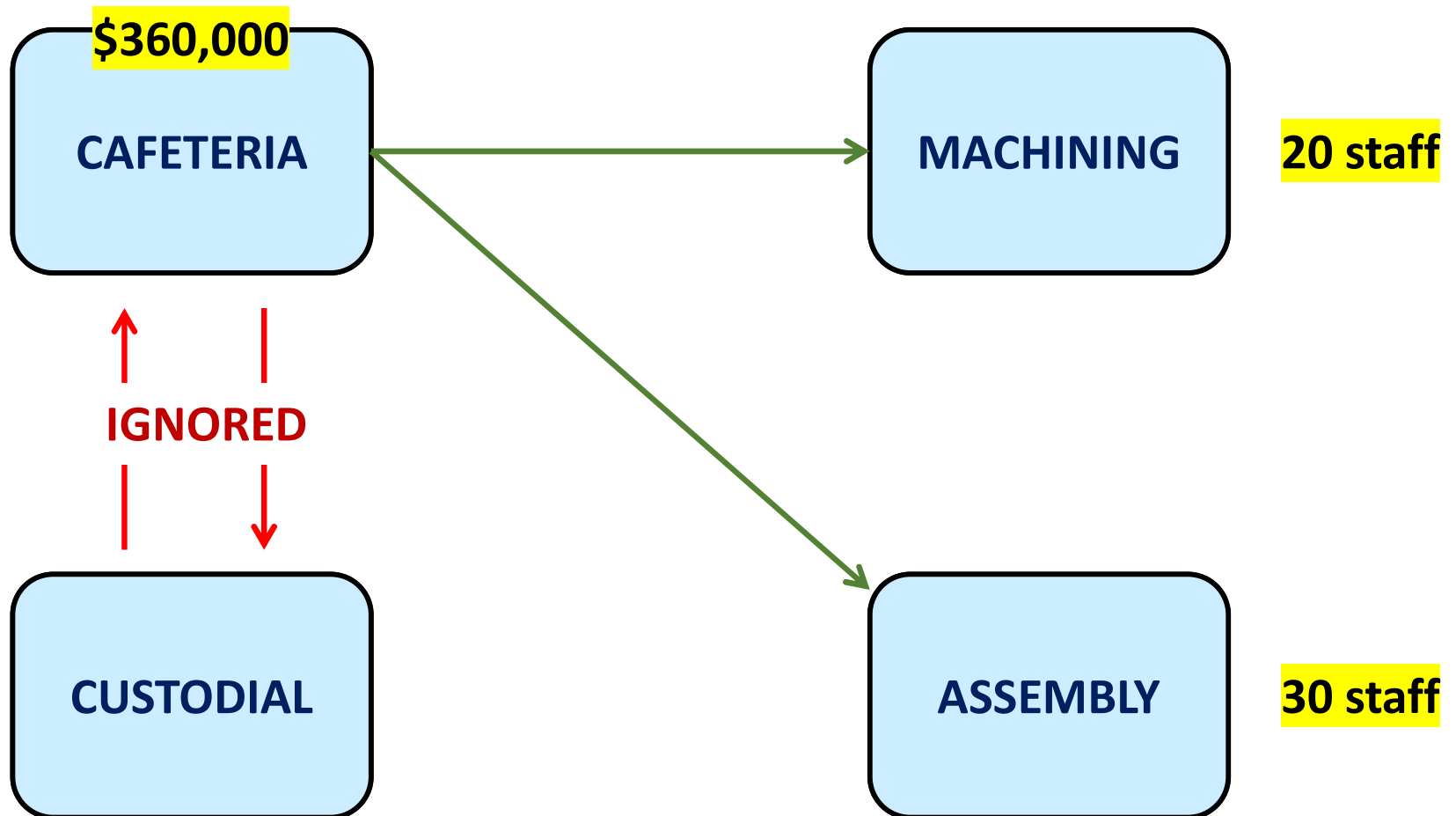
Direct Method – Example

Interactions between service departments are ignored and all costs are allocated directly to operating departments.



Direct Method – Allocate Cafeteria costs

Cafeteria department's allocation base: Number of employees



Direct Method – Allocate Cafeteria costs

Cafeteria department's allocation base: Number of employees

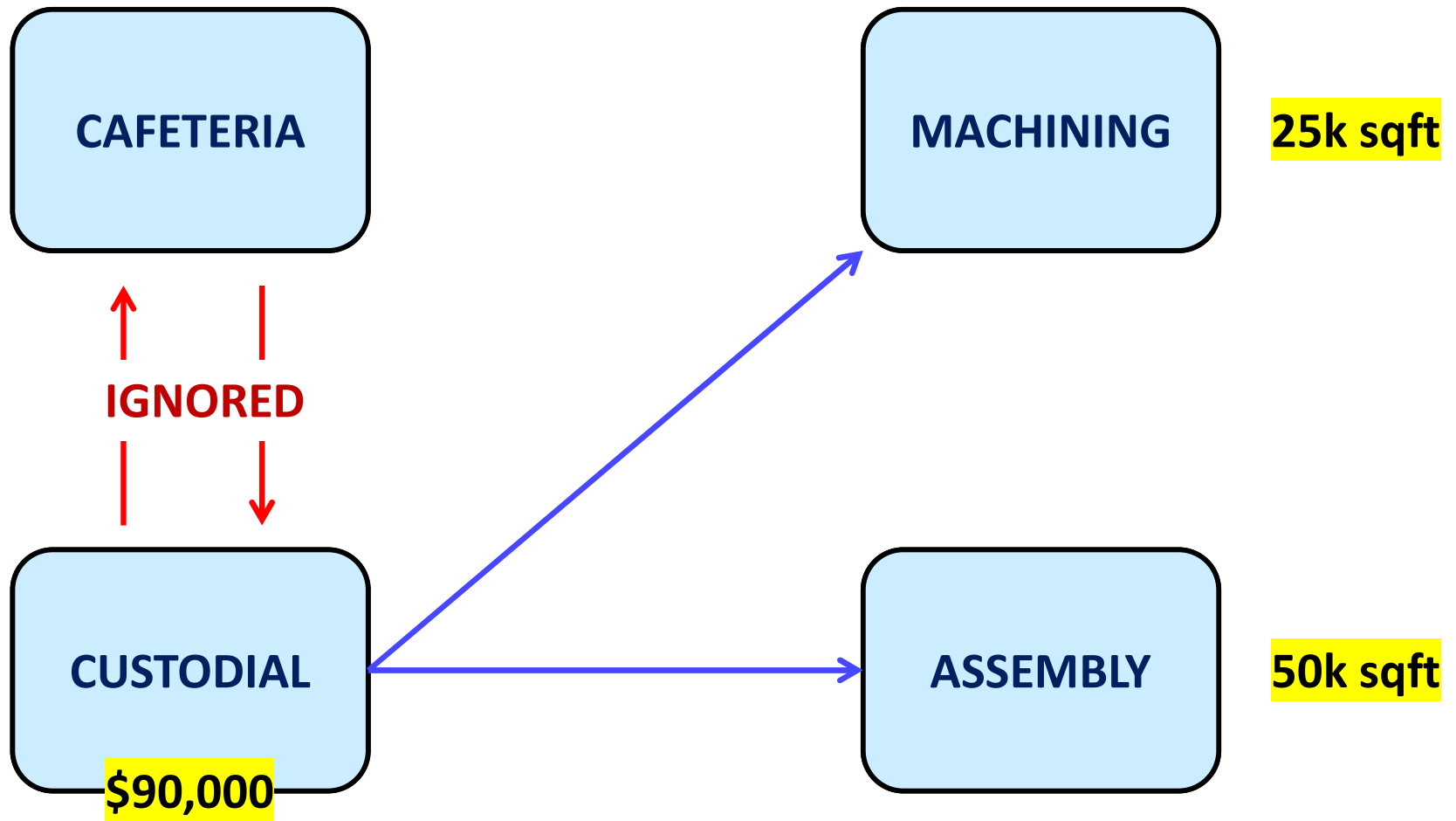
	<u>Service Departments</u>		<u>Operating Departments</u>	
	<u>Cafeteria</u>	<u>Custodial</u>	<u>Machining</u>	<u>Assembly</u>
Departmental costs before allocation	\$ 360,000	\$ 90,000	\$ 400,000	\$ 700,000
Cafeteria allocation	(360,000)		144,000	216,000
Custodial allocation		?	?	?
Total after allocation	\$ 0	?	?	?

$$\$360,000 \times \frac{20}{20 + 30} = \$144,000$$

$$\$360,000 \times \frac{30}{20 + 30} = \$216,000$$

Direct Method – Allocate Custodial costs

Custodial department's allocation base: Square feet occupied



Direct Method – Allocate Custodial costs

Custodial department's allocation base: Square feet occupied

	<u>Service Departments</u>		<u>Operating Departments</u>	
	<u>Cafeteria</u>	<u>Custodial</u>	<u>Machining</u>	<u>Assembly</u>
Departmental costs before allocation	\$ 360,000	\$ 90,000	\$ 400,000	\$ 700,000
Cafeteria allocation	(360,000)		144,000	216,000
Custodial allocation		(90,000)	30,000	60,000
Total after allocation	<u>\$ 0</u>	<u>\$ 0</u>	<u>\$ 574,000</u>	<u>\$ 976,000</u>

$$\$90,000 \times \frac{25k}{25k + 50k} = \$30,000$$

$$\$00,000 \times \frac{50k}{25k + 50k} = \$60,000$$

Direct Method – Summary

After allocation, service departments costs should be fully allocated out.

	<u>Service Departments</u>		<u>Operating Departments</u>	
	<u>Cafeteria</u>	<u>Custodial</u>	<u>Machining</u>	<u>Assembly</u>
Departmental costs before allocation	\$ 360,000	\$ 90,000	\$ 400,000	\$ 700,000
Cafeteria allocation	(360,000)		144,000	216,000
Custodial allocation		(90,000)	30,000	60,000
Total after allocation	<u>\$ 0</u>	<u>\$ 0</u>	<u>\$ 574,000</u>	<u>\$ 976,000</u>

Total is the same

Quick Check 5B.3

	<u>Service Departments</u>		<u>Operating Departments</u>	
	<u>ADMIN</u>	<u>IT</u>	<u>Accounting</u>	<u>Others</u>
Departmental costs before allocation	\$ 180,000	\$ 90,000	\$ 190,000	\$ 900,000
Number of employees	15	5	20	80
Number of PCs	12	20	18	102

Allocation bases:

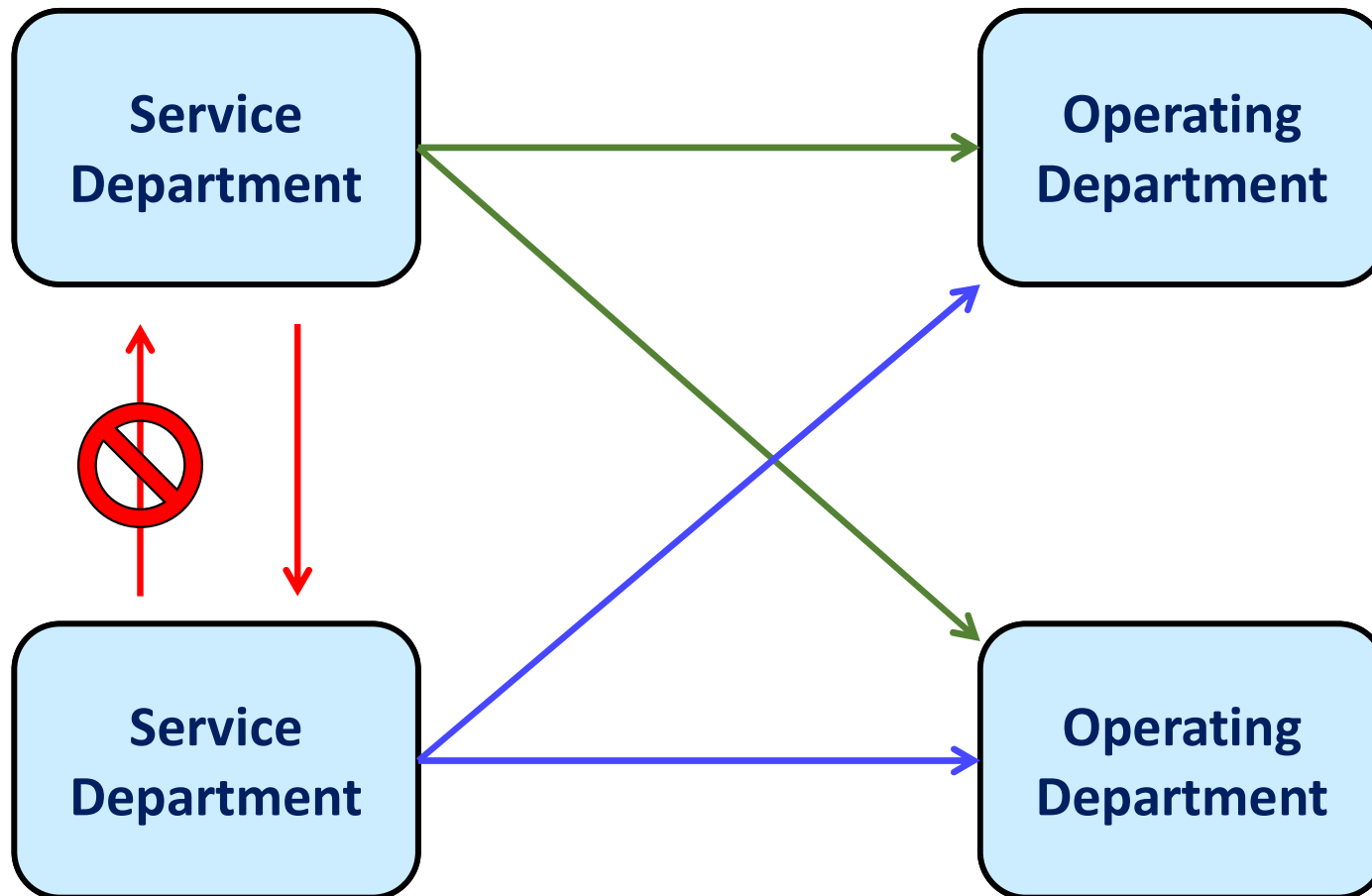
- ADMIN dept: **# of employees**
- IT dept: **# of personal computers**

How much total cost will be allocated from IT to Other using Direct Method?

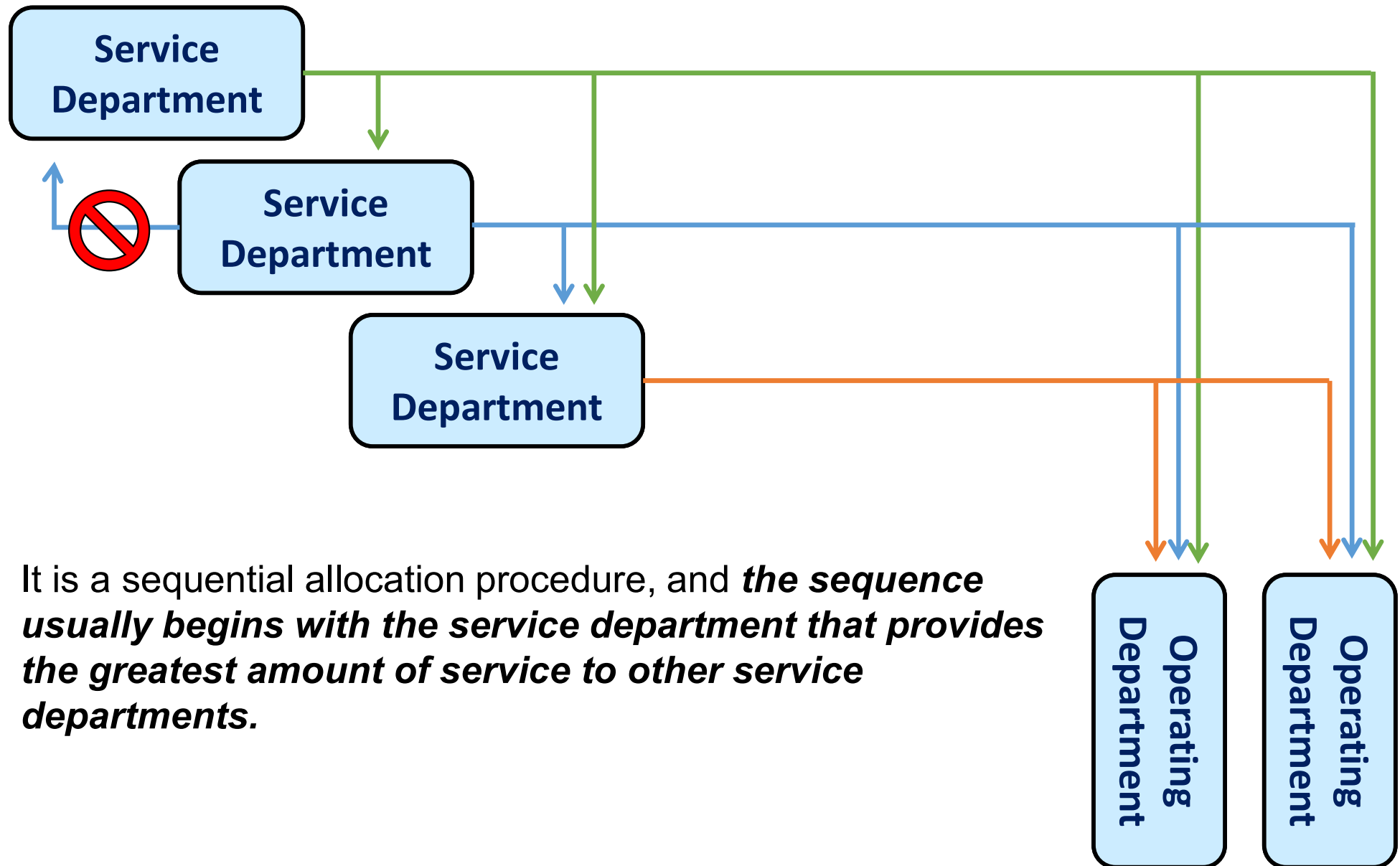
- \$ 69,545
- \$ 60,395
- \$144,000
- \$ 76,500

Step-down Method

Once a service department's costs are allocated, other service department costs are not allocated back to it.



Step-down Method



It is a sequential allocation procedure, and ***the sequence usually begins with the service department that provides the greatest amount of service to other service departments.***

Step-down Method – Example

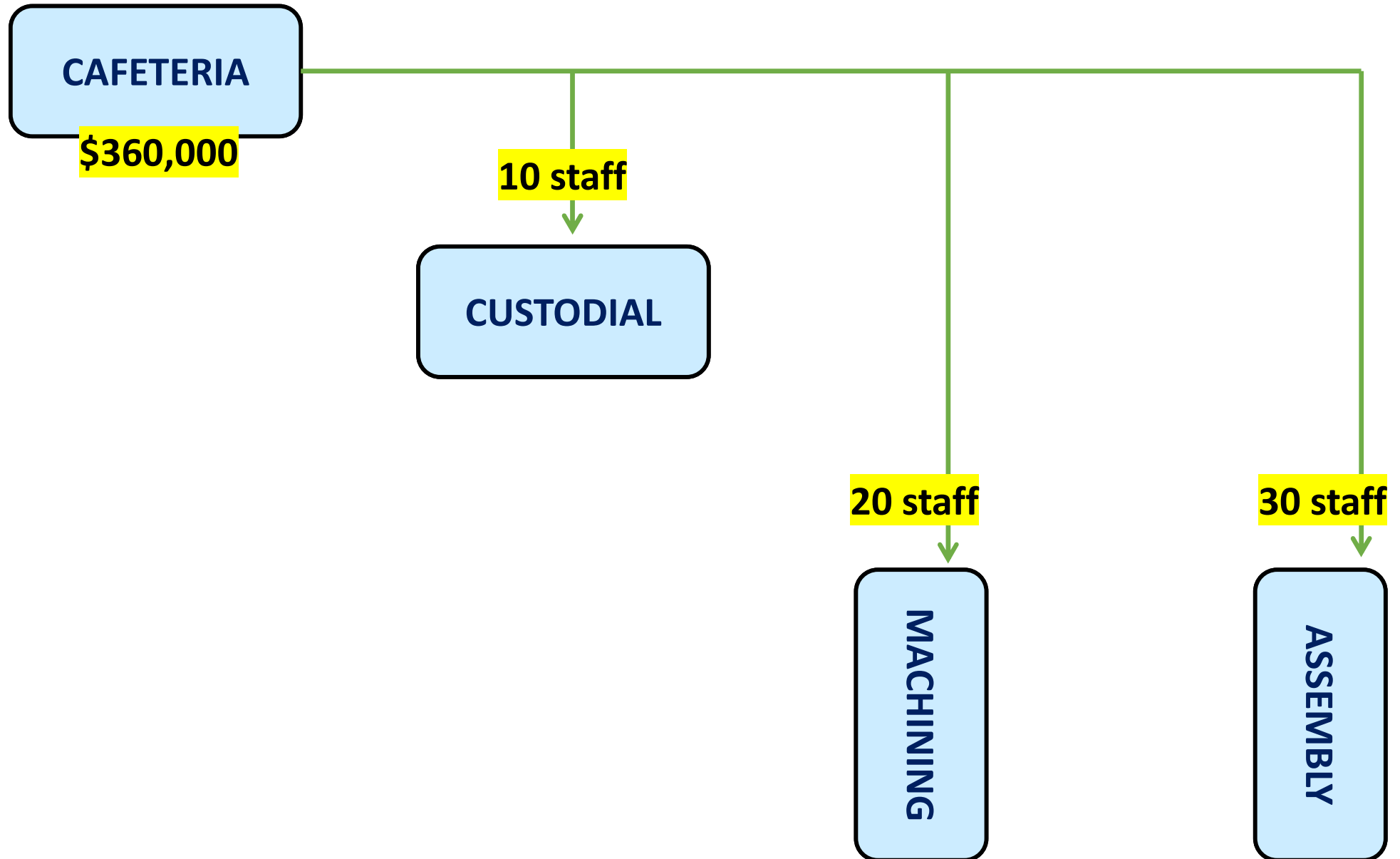
How much of the Cafeteria and Custodial costs should be allocated to each operating department using the direct method of cost allocation?

	<u>Service Departments</u>		<u>Operating Departments</u>	
	<u>Cafeteria</u>	<u>Custodial</u>	<u>Machining</u>	<u>Assembly</u>
Departmental costs before allocation	\$ 360,000	\$ 90,000	\$ 400,000	\$ 700,000
Number of employees	15	10	20	30
Square feet occupied	5,000	2,000	25,000	50,000

<u>Service Department</u>	<u>Allocation Base</u>
Cafeteria	Number of employees
Custodial	Square feet occupied

Cafeteria costs to be allocated first.

Step-down Method – Allocate Cafeteria costs



Step-down Method – Allocate Cafeteria costs

Cafeteria department's allocation base: Number of employees

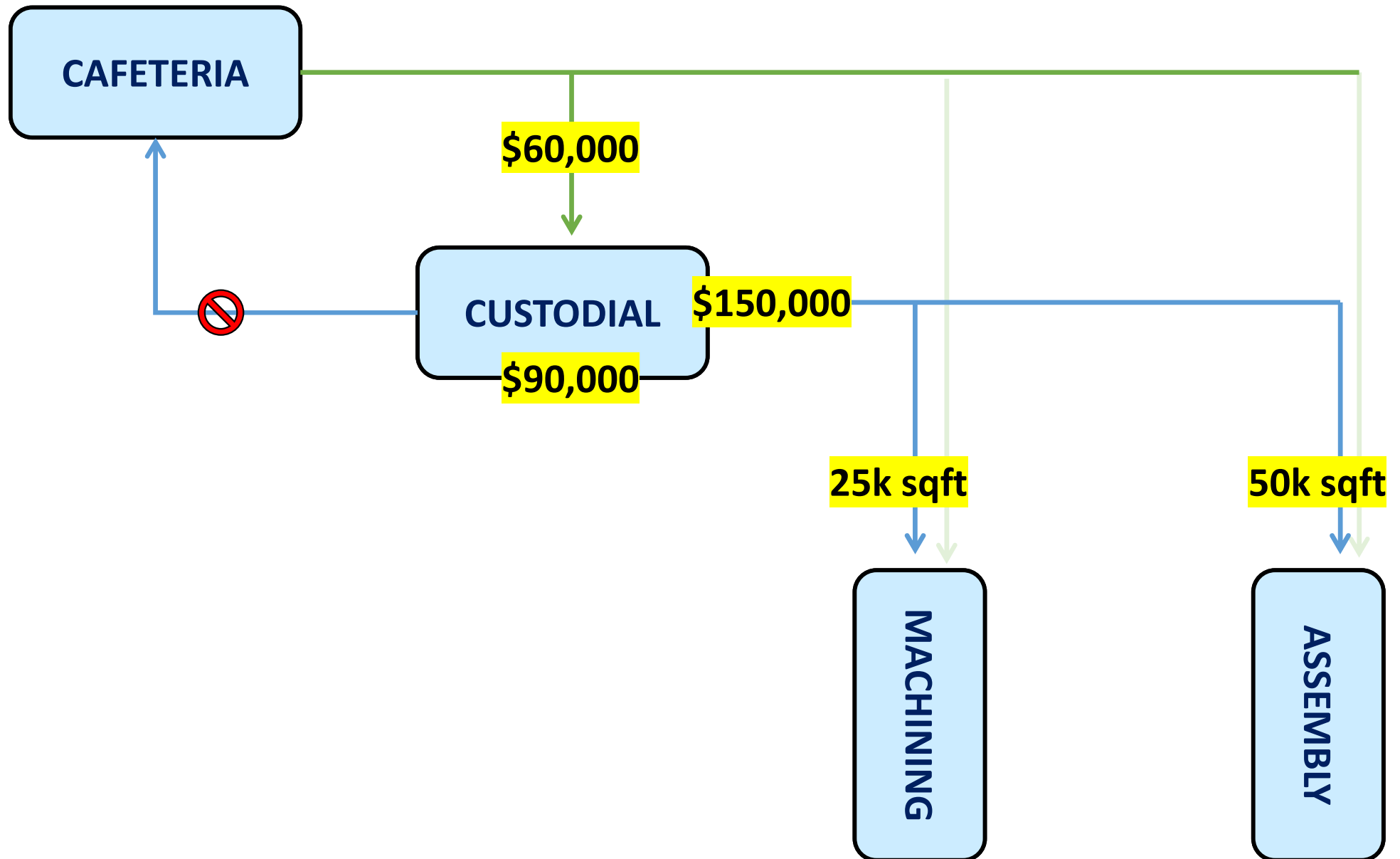
	Service Departments		Operating Departments	
	Cafeteria	Custodial	Machining	Assembly
Departmental costs before allocation	\$ 360,000	\$ 90,000	\$ 400,000	\$ 700,000
Cafeteria allocation	(360,000)	60,000	120,000	180,000
Custodial allocation		?	?	?
Total after allocation	\$ 0	?	?	?

$$\$360,000 \times \frac{10}{10 + 20 + 30}$$

$$\$360,000 \times \frac{20}{10 + 20 + 30}$$

$$\$360,000 \times \frac{30}{10 + 20 + 30}$$

Step-down Method – Allocate Custodial costs



Step-down Method – Allocate Custodial costs

Custodial department's allocation base: Square feet occupied

	Service Departments		Operating Departments	
	Cafeteria	Custodial	Machining	Assembly
Departmental costs before allocation	\$ 360,000	\$ 90,000	\$ 400,000	\$ 700,000
Cafeteria allocation	(360,000)	60,000	120,000	180,000
Custodial allocation		(150,000)	50,000	100,000
Total after allocation	<u>\$ 0</u>	<u>\$ 0</u>	<u>\$ 570,000</u>	<u>\$ 980,000</u>

New total = \$150,000

\$90,000 original Custodial cost
plus \$60,000 allocated from the Cafeteria.

$$\$150,000 \times \frac{25k}{25k + 50k}$$

$$\$150,000 \times \frac{50k}{25k + 50k}$$

Step-down Method – Summary

After allocation, service departments costs should be fully allocated out.

	Service Departments		Operating Departments	
	Cafeteria	Custodial	Machining	Assembly
Departmental costs before allocation	\$ 360,000	\$ 90,000	\$ 400,000	\$ 700,000
Cafeteria allocation	(360,000)	60,000	120,000	180,000
Custodial allocation		(150,000)	50,000	100,000
Total after allocation	<u>\$ 0</u>	<u>\$ 0</u>	<u>\$ 570,000</u>	<u>\$ 980,000</u>

Total is the same

Allocation using Direct and Step-down Method – Key Points

In both the direct and step-down methods,

- Any amount of the allocation base attributable to the service department whose cost is being allocated is always ignored.

In the step-down method,

- Any amount of the allocation base that is attributable to a service department whose cost has already been allocated is ignored.
- Each service department assigns its own costs to operating departments plus the costs that have been allocated to it from other service departments.

Quick Check 5B.4

	<u>Service Departments</u>		<u>Operating Departments</u>	
	<u>ADMIN</u>	<u>IT</u>	<u>Accounting</u>	<u>Others</u>
Departmental costs before allocation	\$ 180,000	\$ 90,000	\$ 190,000	\$ 900,000
Number of employees	15	5	20	80
Number of PCs	12	20	18	102

Allocation bases:

- ADMIN dept: **# of employees**
- IT dept: **# of personal computers**

Allocate ADMIN costs first. Use Step-down method.

How much cost will be allocated from ADMIN to Accounting?

- \$ 36,000
- \$ 30,000
- \$ 34,286
- \$ 20,000

Quick Check 5B.5

	<u>Service Departments</u>		<u>Operating Departments</u>	
	<u>ADMIN</u>	<u>IT</u>	<u>Accounting</u>	<u>Others</u>
Departmental costs before allocation	\$ 180,000	\$ 90,000	\$ 190,000	\$ 900,000
Number of employees	15	5	20	80
Number of PCs	12	20	18	102

Allocation bases:

- ADMIN dept: **# of employees**
- IT dept: **# of personal computers**

Allocate ADMIN costs first. Use Step-down method.

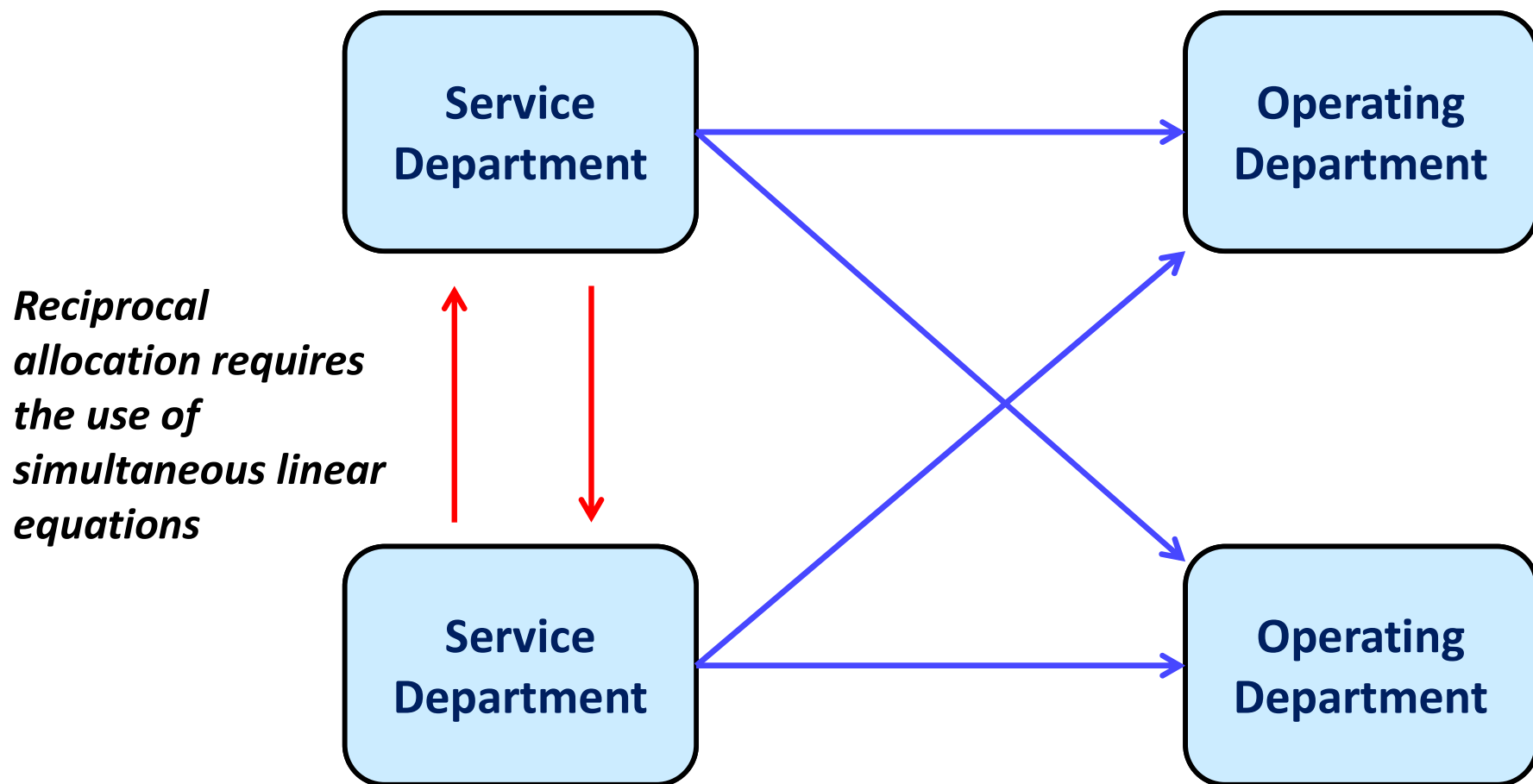
When IT Dept cost is allocated, how much cost needs to be allocated out?

- \$90,000
- \$98,571
- \$97,500
- \$99,000

Reciprocal Method [Excluded from assessment]

Interdepartmental services are given full recognition rather than partial recognition as with the step method.

Because of its mathematical complexity, the reciprocal method is rarely used.



End of topic